ServiceSetu : A Real-Time Home Services Marketplace with Intelligent Booking and Provider Management

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***Abstract* —The rapid growth of urban populations has significantly increased the demand for reliable and efficient home services such as plumbing, electrical repairs, and domestic assistance. However, traditional methods of accessing such services are often unstructured, unreliable, and lack transparency. This paper presents ServiceSetu, a real-time home-services marketplace platform designed to connect customers with verified local service providers through a structured and intelligent booking system. The platform integrates modern web technologies, including a Node.js backend, React-based frontend, and real-time communication using WebSockets, to ensure seamless coordination between users and providers. It incorporates role-based access, geolocation-based discovery, and scalable APIs to enhance efficiency and trust. The system aims to improve service accessibility, ensure transparency, and provide a smooth user experience through real-time updates and optimized workflows.**

Keywords — Home Services, Real-Time Systems, Marketplace Platform, Node.js, React, Booking System, Geolocation

I. INTRODUCTION

The increasing pace of urbanization has led to a growing demand for convenient and reliable home services. Individuals often struggle to find trustworthy professionals such as electricians, plumbers, and domestic helpers through traditional offline methods. These methods rely heavily on personal references and lack consistency in service quality, availability, and pricing.

Digital platforms have attempted to address this gap by introducing online service marketplaces. However, many of these platforms still face issues related to delayed communication, lack of real-time tracking, and inefficient coordination between users and service providers.

ServiceSetu is proposed as a solution to these challenges by offering a structured and scalable platform that enables real-time booking, provider discovery, and service tracking. The system focuses on improving user trust, enhancing operational efficiency, and ensuring seamless interaction between all stakeholders.

II. PROBLEM STATEMENT

The current home-services ecosystem suffers from multiple inefficiencies that affect both customers and service providers. The absence of a centralized and reliable platform leads to difficulty in discovering nearby providers and verifying their authenticity. Moreover, the lack of real-time updates results in poor coordination and uncertainty during the booking process.

Existing platforms often fail to provide a complete solution due to limited workflow management and weak system integration. The key challenges can be summarized as follows:

- Lack of a trusted and centralized platform
- Inefficient provider discovery and verification
- Absence of real-time booking updates
- Poor coordination between users and providers
- Limited transparency in pricing and service completion

These issues highlight the need for a scalable and intelligent system that ensures reliability, transparency, and real-time interaction.

III. OBJECTIVES

The core challenge this work addresses is the lack of a unified, efficient, and user-friendly platform for accessing and managing hyperlocal services. Specifically, the following problems exist in the current landscape:

• Accessing local services often involves fragmented and unorganized methods, such as personal contacts or multiple apps, making the process time-consuming and inconvenient.

• The majority of existing service platforms lack transparency in terms of service quality, pricing, and provider reliability, leading to reduced user trust.

• Many platforms do not ensure proper verification of service providers, increasing the risk of fraud or poor-quality service.

• Booking services through traditional methods is often slow and inefficient, involving manual coordination and lack of real-time availability.

• Existing solutions provide limited or no integrated features such as secure payments, service tracking, and centralized management.

• There is a lack of a standardized rating and feedback system, making it difficult for users to make informed decisions.

• Service providers often lack digital tools to manage bookings, schedules, and customer interactions efficiently.

• Current systems do not fully digitize hyperlocal service management, resulting in operational inefficiencies and lack of scalability.

• Users face difficulty in comparing multiple service providers based on ratings, pricing, and availability in one place.

• The absence of a seamless and intuitive user interface in many platforms reduces accessibility, especially for non-technical users.

IV. LITERATURE SURVEY

The rapid growth of digital platforms has significantly transformed the way users access hyperlocal services. Several existing systems, such as Urban Company, Justdial, and Housejoy, aim to connect users with service providers and offer basic functionalities like service discovery and booking. These platforms utilize features such as geolocation, user reviews, and categorized listings to enhance user experience. However, despite these advancements, most systems focus only on specific aspects of service delivery and lack a fully integrated approach. Limitations such as absence of real-time synchronization, inefficient communication, limited transparency in service tracking, and scalability issues are commonly observed. Furthermore, inconsistencies in workflow management and incomplete implementation of verification and payment systems reduce overall reliability. To address these challenges, the proposed system, ServiceSetu, introduces a comprehensive solution that integrates real-time communication, structured booking workflows, and a scalable modular architecture, ensuring improved efficiency, transparency, and user trust.

In terms of booking functionality, Urban Company offers a complete booking system, Justdial does not support direct booking, Housejoy provides basic booking features, while ServiceSetu implements a structured and systematic booking process. For real-time tracking, Urban Company and Housejoy offer limited capabilities, Justdial lacks this feature entirely, whereas ServiceSetu provides full tracking with continuous status updates. Communication

features are limited in Urban Company and Housejoy and absent in Justdial, while ServiceSetu enables real-time communication between users and service providers. Regarding verification, Urban Company ensures verified professionals, Justdial has limited verification, Housejoy performs only basic checks, and ServiceSetu incorporates a strong verification mechanism. Finally, in terms of platform integration, Urban Company and Housejoy offer moderate integration, Justdial has low integration as it primarily acts as a directory, and ServiceSetu delivers high integration by combining booking, tracking, communication, and management features within a unified system.

Table 1: Comparative Analysis of Home Service Platforms and Applications

No	Work	Yr	Key Features	Limits
1	Web Service	2020	List, Register	No booking/tracking
2	ML App	2023	Chatbot, UPI	No sync
3	HomeWorks	2023	Mobile UX	No tracking
4	Fullstack	2023	Admin, Feedback	No scalability
5	Booking Plat.	2023	Booking, Pay	No tracking
6	UrbanClap Study	2023	Aggregator, Analytics	Quality, fees
7	SignalR App	2024	Real-time chat	No booking

Existing research on online service booking systems generally focuses on specific functionalities rather than a complete end-to-end solution. Most proposed systems support basic features such as service listing, registration, booking, and payment integration. Some studies also include chatbot assistance, mobile-friendly interfaces, or simple admin dashboards; however, they often lack critical functionalities like real-time service tracking, scalability handling, and integrated communication between users and service providers. In many cases, research prototypes are limited to demonstrating individual modules rather than a fully deployed system.

In comparison, ServiceSetu is designed as a more comprehensive and unified platform that integrates all major components of a service ecosystem. It supports user registration, service provider management, real-time booking, secure payment processing, tracking of service status, and feedback collection within a single system. Unlike most existing research models that operate in isolated modules or lack synchronization, ServiceSetu focuses on end-to-end workflow integration with better scalability and real-time updates. This makes it more suitable for practical deployment compared to traditional research-based systems, which often remain at a prototype or limited-functional level.

V. METHODOLOGY

A. Proposed System

The proposed system, **ServiceSetu**, is designed as a comprehensive hyperlocal service marketplace that enables efficient interaction between customers and service providers through a unified digital platform. The system focuses on improving accessibility, transparency, and reliability in service booking and management.

The platform supports three primary user roles: customers, service providers, and administrators. Customers can register, log in, and search for nearby service providers based on service category and location. The system leverages geolocation techniques to identify and display providers within a specified radius, ensuring relevance and convenience.

A structured booking workflow is implemented to streamline the service lifecycle. Customers can create bookings by specifying service details such as date, time, and address. The system validates booking requests to prevent scheduling conflicts and ensures provider availability. Once a booking is created, service providers receive notifications and can accept or reject requests. Upon acceptance, the service progresses through defined states such as pending, accepted, and completed.

Real-time synchronization is a key feature of the proposed system. Using WebSocket-based communication, booking updates are instantly reflected on both customer and provider dashboards. This eliminates delays and ensures consistent system state across all users.

Service providers are equipped with dedicated dashboards that allow them to manage bookings, update service status, track earnings, and configure payout details. The platform also incorporates a review and rating mechanism where customers can provide feedback after service completion. This enhances accountability and helps maintain service quality.

Additionally, the system includes backend support for secure payment processing, order management,

and OTP-based verification for service completion. Even though the frontend payment interface may be optional or disabled in certain deployment modes, the backend architecture is designed to support complete transaction workflows.

Overall, the proposed system overcomes limitations of existing platforms by providing a fully integrated solution that combines real-time updates, structured workflows, secure interactions, and scalable service management.

B. System Architecture

The system architecture of ServiceSetu is designed using a layered client-server model to ensure scalability, modularity, and efficient data flow. The architecture consists of multiple interconnected layers, each responsible for specific functionalities, along with real-time communication and external service integrations.

The system operates through continuous interaction between the client and server, enabling dynamic data exchange, real-time updates, and scalable service management.

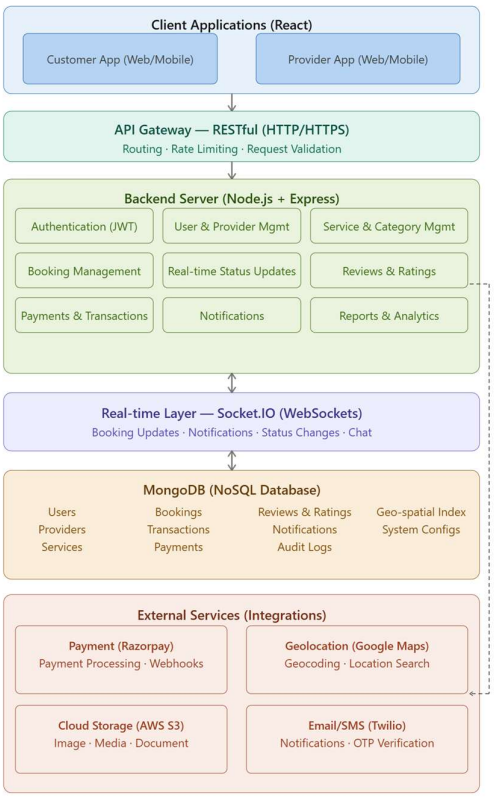

Fig 1. System Architecture

The Presentation Layer comprises the React-based user interface, which provides interactive screens for customers, service providers, and administrators. It includes modules for authentication, service browsing, booking, and dashboard management. This layer handles user inputs, displays dynamic data, and communicates with the backend through RESTful APIs while maintaining application state and session context.

The Business Logic Layer is implemented using Node.js and Express, acting as the core processing unit of the system. It manages authentication using JWT, enforces role-based access control, and handles all major functionalities such as provider onboarding, booking lifecycle management, service validation, and review processing. The backend follows a modular structure with separate routes, controllers, and models, ensuring maintainability and scalability. It also integrates middleware for request validation, security, and error handling.

The Data Layer is responsible for persistent data storage using MongoDB with Mongoose. It stores structured data including user profiles, service provider details, bookings, reviews, and service categories. Geospatial indexing is implemented to support efficient location-based queries for nearby service discovery. This layer ensures data consistency, integrity, and efficient retrieval for all system operations.

The Real-Time Communication Layer enables instantaneous synchronization between users and service providers using Socket.IO. It listens for booking-related events such as creation, acceptance, rejection, and completion, and broadcasts updates to relevant clients. This ensures that both parties have a consistent and up-to-date view of the booking status, significantly improving responsiveness and user experience.

The External Integration Layer connects the system with third-party services to extend functionality. Payment processing is supported through a payment gateway for order creation, verification, and transaction handling. Cloud-based storage services are used for managing uploaded images, while geocoding services convert user-provided addresses into geographic coordinates for location-based features. These integrations enhance the overall capability and reliability of the platform.

The overall system workflow is achieved through seamless coordination between all layers. The frontend initiates requests, the backend processes business logic and interacts with the database, and

real-time events ensure synchronization across users. External services are invoked when necessary, enabling features such as payments, image handling, and geolocation. This layered architecture ensures scalability, modularity, and efficient performance of the ServiceSetu platform.

VI. RESULTS & DISCUSSION

A. User Interface (UI)

The user interface of ServiceSetu was designed to be simple, responsive, and user-friendly for all types of users, including customers and service providers. The React-based frontend provides clearly structured pages such as login/signup, provider listing, booking interface, and dashboards.

The provider listing interface successfully displays available service providers along with relevant details, enabling users to make informed decisions. The booking page allows users to easily select service details such as date, time, and address, ensuring a smooth booking experience. Dashboards for both users and providers present real-time updates of booking status, improving transparency and usability.

The integration of real-time updates enhances the UI experience by instantly reflecting changes without requiring manual refresh. However, minor UI limitations were observed, such as improper rendering of location data and the absence of an active payment interface on the frontend. Despite these issues, the overall interface remains intuitive and efficient for performing core operations.

Fig 2. Landing Page

This screen shows the main landing page of the ServiceSetu platform. It features a prominent headline, “Find Professional Services at Your Doorstep,” along with a short description emphasizing trusted and affordable services. Users can quickly search for services by entering the required service type and location. The top navigation bar includes options like Home, Services, About Us, Contact, and user authentication (Login/Sign Up), providing easy access to different sections of the platform.

Fig 3. Signup Page

This screen displays the user registration page where new users can create an account. It includes input fields for full name, email, phone number, address, and password. The page is designed to be simple and user-friendly, encouraging users to join the platform. A motivational message on the left highlights the benefits of using the service, such as improved lifestyle and access to trusted professionals.

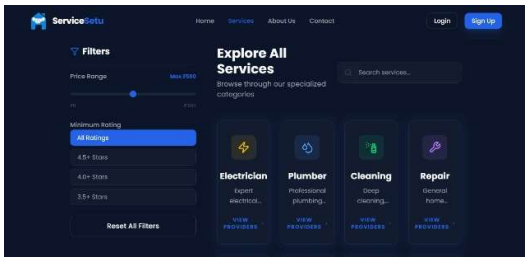

Fig 4. Services Page

This screen presents the service browsing interface. Users can explore various service categories such as electrician, plumber, cleaning, and repair. A filter panel on the left allows users to refine results based on price range and ratings. The main section displays service cards with options to view available providers. A search bar is also included to quickly find specific services, making the platform efficient and easy to navigate.

B. System Performance and Functional Description

The system was evaluated based on its functionality, performance, and reliability across various modules. The booking workflow operates efficiently, supporting all stages from booking creation to completion with proper validation and state management. Real-time synchronization using Socket.IO ensures that updates are instantly communicated between users and providers, reducing delays and improving coordination.

The provider discovery module effectively utilizes geolocation to fetch nearby service providers,

ensuring relevant results and faster access. The role-based access control system successfully restricts functionalities based on user roles, enhancing security and maintaining system integrity.

The backend architecture demonstrates strong performance due to its modular design and efficient API handling. The system is capable of handling concurrent requests while maintaining data consistency. Additionally, features such as review and rating mechanisms contribute to improved service quality and user trust.

Despite these strengths, certain challenges were identified. The mismatch between backend payment capabilities and frontend implementation creates a gap in the user experience. Redundant booking routes and multiple completion flows may lead to inconsistencies if not optimized. Furthermore, the lack of comprehensive automated testing limits the ability to ensure long-term reliability.

Overall, the system performs effectively in delivering a real-time, scalable, and integrated solution for hyperlocal services, while highlighting areas for future improvement.

VII. CONCLUSION

The proposed system, ServiceSetu, successfully addresses the challenges associated with traditional and existing hyperlocal service platforms by providing a unified, efficient, and real-time solution. The system integrates essential functionalities such as service discovery, structured booking workflows, role-based access control, and real-time communication into a single platform.

The implementation demonstrates that a modular client-server architecture combined with modern web technologies can significantly improve service accessibility, transparency, and reliability. Features such as geolocation-based provider discovery, real-time booking updates, and a review system enhance user experience and build trust among users and service providers.

Although certain limitations exist, such as the temporary absence of frontend payment integration and minor system inconsistencies, the overall system achieves its primary objectives effectively. The architecture is scalable and maintainable, allowing future enhancements without major structural changes.

In conclusion, ServiceSetu provides a robust foundation for a next-generation hyperlocal service marketplace, offering improved efficiency, better user interaction, and a reliable digital ecosystem for service management.

VIII. FUTURE SCOPE

- Integration of a complete frontend payment system to align with backend capabilities and enable seamless transactions.
- Implementation of advanced search and filtering options using AI-based recommendations for better service matching.
- Development of a mobile application to increase accessibility and user engagement.
- Addition of real-time chat functionality between users and service providers.
- Enhancement of security features such as two-factor authentication and improved data encryption mechanisms.
- Integration of analytics and reporting tools for providers and administrators to track performance and usage trends.
- Optimization of system architecture to support large-scale deployments and high user traffic.
- Implementation of automated testing frameworks and CI/CD pipelines for improved reliability and maintainability.
- Support for multi-language interfaces to expand usability across diverse user groups.

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